

BSc Computer Systems Engineering Project Proposal

SafeSurf AI



Submitted By

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1: Introduction

1.1 : Overview

With the rapid growth of the internet, users face several hidden risks such as harmful websites, phishing emails, and unsafe public Wi-Fi networks. These threats are not realized until it is too late by most people. Current tools tend to be disjointed and ineffective in their speed or the number of security checks they attain. This project will address this problem by developing one real time platform which will monitor all the three main sources of risks like websites, emails, and Wi-Fi networks. Its system assists users to be on the safe side, as in the form of displaying instant alerts, safety scores and threat reports. It helps users against being suspicious websites, data theft and unsafe browsing without the need to be tech-savvy.

1.2 : Objectives

- To design a simple and smart system that checks visited websites for security in real time.
- To detect suspicious or phishing emails automatically and analyze the links inside them.
- To evaluate Wi-Fi networks and identify doubtful or risky public hotspots.
- To provide a joint dashboard that shows risk scores, alerts, and clear reports.
- To assist users in making safe online decisions through live notifications and warnings.

1.3 : Literature Review

The growing amount of digital communication, online browsing, and wireless connectivity has significantly increased users' experience to cyber threats. These threats mainly create from phishing websites, malevolent emails, and insecure Wi-Fi networks, each showing severe risks to user privacy and data security [1]. Although research in cybersecurity has progressive across these distinct domains, most existing systems examine these threats in isolation, resulting in limited real-time security and reduced effectiveness for end users [2]. The requirement of multi-layered security frameworks that are integrated has thus taken precedence causing the creation of combined systems that could easily analyze the URL threat, email threat, and Wi-Fi threat simultaneously [3].

The field of phishing URL detection has also achieved significant advancement because of the use of deep learning and advanced sequence modeling algorithms. Haque et al. developed a 1D-CNN network that can process both lexical and structural URLs patterns and have high

detection rates of phishing links that have been generated recently [1]. To widen this path, Sruthi utilized a Siamese LSTM model that recognizes mimic-domain phishing URLs through the learning of relational similarities between legit and malicious domains [2]. Other innovations are predictive blacklisting algorithms that can represent the URLs as featureful [3] before widely known, CNN-based URLNet models to learn URL representation features [4], and transformer-based systems utilizing BERT embeddings to learn contextual features of URLs [5]. Although these solutions have been made, they are still constrained by offline assessment, computational costs,

Phishing attacks via email still pose some new challenges, and researchers have to investigate the deep learning-based defense features and NLP-based defenses. Altwaijry et al. resorted to the hybrid CNN-GRU model that is effective at capturing both sequential and spatial cues in email text, which results in a higher detection accuracy compared to the conventional machine learning procedures. [6]. There are other methods, which use CNNLSTM hybrid networks to better represent long email sequences [7] and BERT-based models that incorporate deep contextual understanding for improved classification robustness [8]. TF-IDF with SVM classifiers are also lightweight NLP methods that have been found useful in fast and computationally efficient email screening. [9].

To offer comprehensive defense against phishing assaults, multi-layer detection systems also examine email headers, message content, and embedded URLs [10]. Nevertheless, these methods typically don't integrate with actual email services like Gmail and don't check links in emails in real time [10].

Parallel advancements in Wi-Fi and wireless intrusion detection show significant progress in identifying unusual network activity. Bhutta et al. introduced a LightGBM-based intrusion detection system that is fast and lightweight for Wi-Fi threat monitoring [11]. Deep learning models like CNN-RNN hybrids have improved detection accuracy by capturing complex spatial and temporal network patterns [12]. Additional research focuses on finding unsafe public Wi-Fi hotspots by examining encryption standards, access point features, and connection reliability with deep learning methods [13]. For IoT environments, quantized DNN-BiLSTM models have been suggested to keep strong accuracy while reducing computing demands, allowing them to work on devices with limited resources [14]. Surveys outlining these advancements highlight the growth of wireless intrusion detection, but they also point out the absence of user-friendly, real-time systems suitable for everyday users

[15]. Most current solutions need technical setup, depend on packet-capture tools, and lack easy-to-use scoring systems or consolidated dashboards. Overall, the literature demonstrates extensive progress in detecting phishing URLs, malicious emails, and wireless intrusions; however, these advancements remain fragmented across separate domains. The absence of unified, real-time, and user-centered solutions continues to limit practical cybersecurity protection for non-expert users [15]. These gaps highlight the necessity of integrated systems such as SafeSurfAI, which aims to combine URL, email, and Wi-Fi threat monitoring into a single, intelligent platform supported by real-time alerts, enriched threat intelligence, and an accessible user interface. Across these studies, the main limitations include:

- Focus on a single threat type instead of multi-layer protection
- Offline testing rather than real-time detection
- No user dashboard or alerting system
- No integration with APIs like VirusTotal, SafeBrowsing, WHOIS, IPinfo, or Gmail API
- No cross-analysis of URL + Email + Wi-Fi threats
- No automatic reporting or permission-based privacy model

The literature shows strong progress in detecting phishing URLs, malicious emails, and network intrusions. However, each system works independently and focuses on one type of threat. Most studies use offline datasets, lack real-time protection, and do not provide user dashboards or alerts. None of the reviewed papers combine URL, email, and Wi-Fi detection, nor do they integrate threat-intelligence APIs, real-time scoring, or automated reporting.

These gaps highlight the need for a system like SafeSurf AI, which unifies web, email, and Wi-Fi security into a single intelligent protection dashboard. SafeSurf AI addresses these limitations by combining website scanning, email phishing detection, and Wi-Fi risk evaluation into one real-time system. It uses external threat-intelligence APIs, monitors browser activity, extracts links from Gmail, and checks Wi-Fi encryption. The system offers live scoring, alerts, maps, and downloadable reports, providing a practical solution for real users. This makes SafeSurf AI more user-focused and ready for use compared to existing research.

Table 1: Comparison of Previous Studies

Ref	Paper Title	Journal Name / Year	Relevance	Methodology	Findings	Limitations
[1]	Detecting Phishing URLs Based on a Deep Learning Approach	Applied Sciences, 2024	Supports URL safety module	1D-CNN on lexical + structural URL features	High accuracy; detects newly generated phishing URLs	No real-time scanning; URL-only
[2]	Phishing URL Detection Using Siamese LSTM Networks	IJCA, 2025	Useful for similarity-based phishing detection	Siamese LSTM comparing URL similarity	Detects mimic-domain attacks	No API integration, offline only
[3]	PhishNet: Predictive Blacklisting for Phishing Detection	IEEE Security & Privacy, 2023	Helps with predictive URL threat detection	ML-based predictive blacklist generation	Detects phishing before blacklists update	Dependent on huge datasets; not real-time for users
[4]	URLNet: Learning URL Representations for Malicious URL Detection	WWW Journal, 2023	Strong deep-learning model for URL analysis	CNN over character + word-level URL inputs	Strong detection accuracy on unseen URLs	High computation cost; offline
[5]	Real-Time Malicious URL Detection Using BERT Embeddings	Computers & Security, 2024	Transformer-based URL scoring	BERT embeddings + classifier	Very high detection accuracy	Requires GPU, less lightweight
[6]	Advancing Phishing Email Detection: Deep Learning Comparison	Sensors, 2024	Supports email phishing module	CNN-GRU hybrid model	Best performance among evaluated models	Offline dataset only; no Gmail link checking
[7]	Hybrid CNN–LSTM Email Phishing Classification	IEEE Access, 2023	Combines sequential + structural features	CNN + LSTM email classifier	High accuracy on long and complex emails	Heavy model; not real-time
[8]	BERT-Based Email Phishing Detection System	Expert Systems With Applications, 2024	Advanced NLP model for email detection	BERT fine-tuned classifier	Excellent precision and recall	Needs high computational power
[9]	Automatic Email Phishing	Information Systems,	Baseline classical ML	TF-IDF + SVM /	Fast and lightweight	Lower accuracy for

	Detection Using NLP & ML	2023	for email threats	Logistic Regression		advanced phishing
[10]	Multi-Layer Email Threat Detection Using URL & Header Features	Computers & Security, 2024	Matches SafeSurfAI (checks URLs inside emails)	Multi-layer model using header + body + URL	Better detection than content-only models	No integration with real email services
[11]	Lightweight Real-Time WiFi-Based Intrusion Detection Using LightGBM	Wireless Networks, 2023	Supports Wi-Fi threat detection	LightGBM classifier on Wi-Fi packet data	Fast & accurate	Needs packet capture; not user-friendly
[12]	Hybrid CNN–RNN Deep Learning Framework for Wireless IDS	IJCNIS, 2025	Strong intrusion detection foundation	CNN + RNN hybrid IDS	High dataset accuracy	Not designed for everyday consumers
[13]	Deep Learning Framework for Securing Public Wi-Fi Hotspots	Journal of Network Security, 2024	Helps Wi-Fi hotspot safety scoring	DL analysis of encryption + AP behavior	Detects unsafe hotspots effectively	No user-focused dashboard
[14]	IoT Intrusion Detection Using Quantized DNN–BiLSTM	JNSA, 2025	Lightweight model supports real-time efficiency	Quantized DNN + BiLSTM	Small model size with good results	Focuses on IoT, not general Wi-Fi
[15]	Survey on ML-Based Wireless Intrusion Detection Systems	ACM Computing Surveys, 2024	Provides complete background for Wi-Fi IDS	Survey of 70+ ML-based IDS models	Highlights gaps & improvements	Survey only; no implementation

2: Methodology

2.1: Design

The design phase focused on creating a user-friendly dashboard with real-time security insights.

- **Dashboard Layout:** Clean, professional interface showing security score, threat summary, website security, system status, and recent alerts.

- **Extension Design:** A browser extension that works in real-time to scan active tabs, block unsafe sites, and show protection scores via pop-up.
- **Multi-Feature Integration:** Separate modules for Website Check, WiFi Security, Email Scanner, Threat History, Reports, and Settings.
- **Automated Updates:** Trusted and blocked lists update every 24 hours using phishing threat intelligence feed and other threat intelligence sources.

2.2: Implementation

The implementation involved integrating multiple APIs and building both frontend and backend systems.

APIs Used:

- **VirusTotal** – for scanning URLs and files.
- **Google Safe Browsing** – to detect phishing sites.
- **WHOIS** – for domain registration details.
- **OpenPhish** – real-time suspicious URL database.

Key Features Built:

- **Real-Time Browser Extension:** Scans active tabs, sends data to dashboard, blocks risky sites, and displays safety score in pop-up.
- **WiFi Scanner:** Auto-detects & scans WiFi networks for vulnerabilities.
- **Email Scanner:** Extracts sender details & checks content for malicious links.
- **Multi-Scan Modes:** Basic, Advanced & Bulk scanning options.
- **Pattern-Based Detection:** Uses URL patterns and trusted/blocked lists updated daily.

Backend System:

- Database connected for storing scan history and reports.
- API key management system (3/4 active as per dashboard).
- Automated threat list refresh every 24 hours.

2.3: Testing

Testing was done to ensure reliability and accuracy.

- **Functional Testing:** Each module (Website, WiFi, Email) tested with safe and malicious samples.
- **Extension Testing:** Verified real-time scanning, pop-up scoring and blocking functionality.

- **API Integration Testing:** Ensured all APIs (VirusTotal, Safe Browsing, etc.) respond correctly and update the dashboard.
- **Performance Testing:** Checked system performance during bulk scans and multi-tab scanning.
- **User Acceptance Testing:** Feedback taken on dashboard clarity and ease of use.

2.4: Evaluation

The system was evaluated based on:

- **Accuracy:** Correctly identifying safe vs. dangerous websites, emails, WiFi networks.
- **Usability:** Intuitive dashboard and easy to understand scoring system.
- **Speed:** Real-time scanning without slowing down browsing.
- **Completeness:** All promised features (Website, WiFi, Email, Extension, Reports) working as intended.
- **Security:** Secure handling of API keys and user data

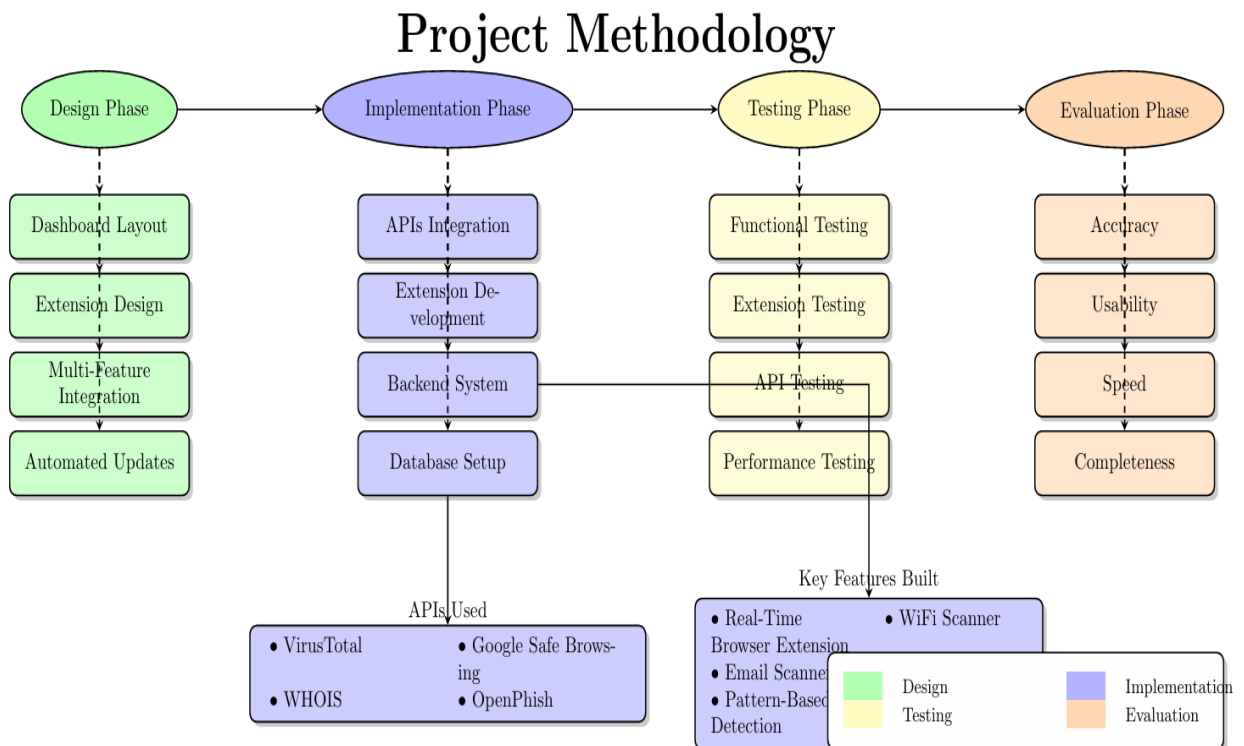


Fig 1: Methodology Flow

3: Project Planning

3.1: Work Schedule Plan

Table 2: Work Schedule Plan

Tasks	Duration
Task 1: Problem Identification & Requirements	Two Weeks
Task 2: Literature Review (URL, Email, Wi-Fi)	Three Weeks
Task 3: System Design(Architecture + Flow + Modules)	Two Weeks
Task 4: Dataset Collection & Preprocessing	One Week
Task 5: Model Development (URL + Email + Wi-Fi Scoring Engine)	One Month
Task 6: API Integration(VirusTotal, SafeBrowsing, Gmail, IPinfo)	Two Weeks
Task 7: Frontend Development(HTML/CSS/JS Dashboard)	One Month
Task 8: Backend Development(Flask/FastAPI + WebSockets)	One Month
Task 9: Real-Time Monitoring + Alerts + Warning Sound	Two Weeks
Task 10: Testing, Debugging & Security Evaluation	Two Weeks
Task 11: Research Paper Writing (IEEE/Elsevier Style)	One Month
Task 12: Final Submission & Viva Preparation	Two Weeks

3.2: Gantt Chart

Table 3: Gantt Chart

Year	2025 to 2026						
Months	M1	M2	M3	M4	M5	M6	M7
Task 1: Problem Identification & Requirements	■						
Task 2: Literature Review (URL, Email, Wi-Fi)	■	■					
Task 3: System Design(Architecture + Flow + Modules)		■	■				
Task 4: Dataset Collection & Preprocessing			■				
Task 5: Model Development (URL + Email + Wi-Fi Scoring Engine)			■	■			
Task 6: API				■	■		

Integration(VirusTotal, SafeBrowsing, Gmail, IPinfo)							
Task 7: Frontend Development(HTML/CSS/JS Dashboard)				■	■		
Task 8: Backend Development(Flask/FastAPI + WebSockets)					■	■	
Task 9: Real-Time Monitoring + Alerts + Warning Sound						■	
Task 10: Testing, Debugging & Security Evaluation						■	■
Task 11: Research Paper Writing (IEEE/Elsevier Style)							■
Task 12: Final Submission & Viva Preparation							■

4: Required Hardware and Software

4.1: Hardware:

- Laptop or PC with internet

4.2: Software:

- Python environment
- Free APIs (VirusTotal, WHOIS, Google Safe Browsing)
- HTML, CSS, JavaScript for dashboard
- SQLite database

5: Budget Description

Zero Budget Project

This project is designed to be completed with zero commercial cost. We will use only free and publicly available resources i.e. open-source software, free APIs, and basic system tools. All development, testing, and deployment will be done using personal computers and current internet access. No paid services, hardware purchases, or external funding are compulsory at any phase.

The complete system will be built using free programming libraries, frameworks, and platforms. Security scans and exploration will be handled through free-tier tools and open-

source clarifications. Every component of the project is planned in a way that avoids any financial expense while still sustaining quality and functionality.

Since all tools and technologies are freely existing, this project proves that a complete and professional system can be established without spending money. The focus is on smart use of capitals, technical skills, and efficient planning rather than financial investment.

6: Sustainable Development Goals

Table 4: Sustainable Development Goals

☐ No Poverty	☐ Zero Hunger
☐ Good Health and Well-Being	☐ Quality Education
☐ Gender Equality	☐ Clean water and Sanitation
☐ Affordable and Clean Energy	☐ Decent Work and Economic growth
☐ Industry, Innovations and Infrastructure	☐ Reduced Inequalities
☐ Sustainable Cities and Communities	☐ Responsible Consumption and Production
☐ Climate action	☐ Life Below Water
☐ Life on Land	☐ Peace, Justice and Strong Institutions
☐ Partnerships	

7: Description of Industrial Support (If any)

N/A

8: Reference

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9: Supervisor's Comments

Supervisor's

Comments: _____

Signature of Supervisor: _____

Name of Supervisor: _____

Designation of Supervisor: _____

Co-Supervisor (If any)

Signature of Co-Supervisor: _____

Name of Co-Supervisor: _____

Designation of Co-Supervisor: _____

Project Serial No.: _____ Dated: _____	Signatures FYDP Committee
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☐ Proposal Approved	☐ Not Approved	☐ Returned for Clarification / Modification
Comments: (if any)		

(Signature of Chairperson)
 Date: _____